

Michał Siemaszko

Curriculum vitae

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Education

- 2021 – 2026 **PhD**, *University of Warsaw*, Poland
Faculty of Mathematics, Informatics and Mechanics
- 2018 – 2021 **Master of Science**, *University of Wrocław*, Poland
Theoretical and teachers physics; grade: *bardzo dobry* (5/5)
- 2015 – 2018 **Bachelor of Science**, *University of Wrocław*, Poland
Theoretical and experimental physics; grade: *bardzo dobry* (5/5)

PhD project

- Research topic Employing multiphoton quantum interference for selected quantum information processing tasks
- Supervisor prof. Magdalena Stobińska
- Description Investigation of a quantum machine algorithm, suitable for an experimental implementation in available photonic quantum computers. Specific focus on the time-series prediction task and experimental implementation of the quantum memristor.

Master's thesis

- Title Strong correlations in open quantum systems
- Supervisor prof. Piotr Ługiewicz
- Description Analysis of linear maps defined on density matrices of open quantum systems (reduced dynamics) with the focus on breaking of complete positivity.

Bachelor's thesis

- Title Investigation of MINERVA experiment data using NuWro Monte Carlo generator with spectral function model
- Supervisor prof. Jan T. Sobczyk
- Description Attempt to understand the discrepancy between physical data and data generated by NuWro (Monte Carlo neutrino event generator). Work was done in CERN ROOT and Mathematica.

Publications

Peer-Reviewed

1. Siemaszko, Michał, Buraczewski, A., Le Saux, B. & Stobińska, M. Rapid training of quantum recurrent neural networks. *Quantum Machine Intelligence* 5, 31 (2023).
2. Bonus, T., Sobczyk, J. T., Siemaszko, Michał & Juszczak, C. Data-based two-body current contribution to the neutrino-nucleus cross section. *Phys. Rev. C* **102**, 015502 (2020).

Preprints

3. Selimović, M., Agresti, I., Siemaszko, Michał, Morris, J., Dakić, B., Albiero, R., Crespi, A., Ceccarelli, F., Osellame, R., Stobińska, M. & Walther, P. *Experimental neuromorphic computing based on quantum memristor* arXiv preprint arXiv:2504.18694.

Experience

Internships

- 2/2022 – 6/2022 **Φ-Lab, ESRIN, Frascati, Italy**
Visiting researcher; development of an quantum machine learning algorithm for time series prediction
- 10/2017 – 9/2019 **University of Wrocław, Wrocław, Poland**
Development of neutrino Monte Carlo event generator – NuWro
- 7/2017 **Institute of Nuclear Physics Polish Academy of Sciences, Kraków, Poland**
Research on Higgs boson and W boson coupling using angle distribution of charged leptons

Conferences

- 11/2025 **Quantum Techniques in Machine Learning (QTML2025), Singapore**
Poster: *Quantum reservoir computing with multiple photonic memristors and skip connections*
- 5/2025 **Quantum Certification Conference+ (QUACC+) , Warsaw, Poland**
Poster: *Time series processing with quantum memristor*
- 6/2024 **Masovian Symposium on Quantum, Optical and Atomic Physics, Warsaw, Poland**
Poster: *Time series processing with quantum memristor*
- 5/2024 **QUANTUMatter24, San Sebastian, Spain**
Poster: *Time series processing with quantum memristor*
- 11/2022 **Quantum Techniques in Machine Learning, Naples, Italy**
Talk: *Rapid training of quantum recurrent neural networks*
- 9/2022 **European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases, Grenoble, France**
Poster: *Rapid training of quantum recurrent neural networks*

Seasonal schools

- 12/2025 **Winter Innovation School, Warsaw, Poland**
- 4/2025 **Quantum Machine Learning: theory meets experiments – Workshop 2025, Vienna, Austria**
- 6/2024 **Summer School on Mid-Infrared Quantum Sensing and Technologies, Warsaw, Poland**
- 7/2023 **Mathematical foundations of Quantum Machine Learning, Trento, Italy**
- 9/2022 **VCQ and AppQInfo Summer School 2022 , Vienna, Austria**
- 7/2022 **Esquisses Summer School, Porquerolles, France**

Talks

- 03/2026 **Quantum Infinity for You, Faculty of Physisc, University of Warsaw**
Talk: *Introduction to the gate-based quantum computing*
- 10/2025 **Quantum Information Seminar, Faculty of Physisc, University of Warsaw**
Talk: *Neuromorphic computing with quantum memristors*
- 11/2022 **Information Theory Seminar, Faculty of Mathematics, Informatics and Mechanics, University of Warsaw**
Talk: *Introduction to the quantum computing*

Teaching

- 10/2022 – 6/2025 **Teacher assistant, University of Warsaw, Warsaw, Poland**
Calculus, probability theory, and linear algebra; 270 teaching hours and over 120 students in total
- 9/2018 – 9/2021 **School teacher, Wrocław, Poland**
Teaching physics, chemistry, and computer science in primary school

Languages

Polish	Native
English	Professional proficiency

Skills

Programming	Python, C++, Mathematica, Bash, HTML, CSS
OS	Linux (Ubuntu, Manjaro, Arch), Windows

Misc	$\LaTeX$, Git, Inkscape, Gnuplot
Packages	NumPy, SciPy, Pandas, PennyLane, StrawberryFields, Piquasso, TensorFlow, PyTorch, Matplotlib

Hobbies

amateur astrophotography / data visualization / philosophy of science / choral singing